GATE ME 90 Days Challenge

Day 66/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Weekly Revision	**Week 11 Revision**	**35 days to GATE**

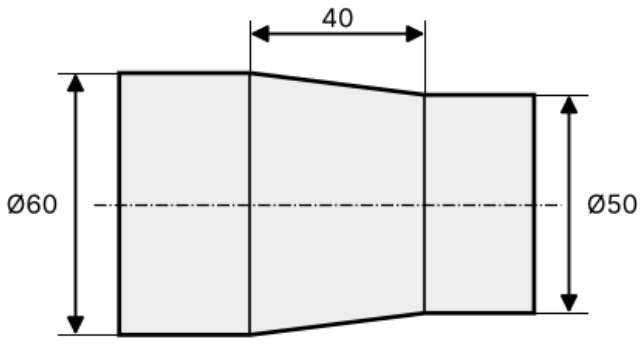
How to use this set: 15 questions (8 MCQ, 7 NAT) on a mixed revision of this week: machine tools, NTM and additive manufacturing (taper turning, milling, SLS), metrology and CIM (CNC constant surface speed, GD&T, CAPP), probability and statistics (geometric and normal distributions, regression lines), production planning and inventory (EOQ, moving averages, SPT) and operations research (AOA networks, M/M/1 queues, shadow prices). Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 NAT

The short taper shown (from Ø60 mm to Ø50 mm over a length of 40 mm) is to be turned on a centre lathe by swivelling the compound rest. The angle (in degrees) through which the compound rest must be swivelled (round off to two decimal places) is _____



Q2 MCQ

Which one of the following additive manufacturing processes fuses a bed of polymer powder layer by layer with a laser beam?

(A) Stereolithography (SLA)
(B) Fused deposition modelling (FDM)
(C) Selective laser sintering (SLS)
(D) Laminated object manufacturing (LOM)

Q3 MCQ

Compared with up (conventional) milling, down (climb) milling

(A) requires a backlash eliminator in the table feed screw because the cutter tends to pull the work into itself

(B) starts each cut at zero chip thickness, causing rubbing

(C) tends to lift the workpiece off the table

(D) is preferred for castings with hard, sandy skin

Q4 NAT

A CNC lathe program uses G96 S180 (constant surface speed of 180 m/min). The spindle speed (in rpm) when the tool is cutting at a diameter of 40 mm (round off to one decimal place) is _____

Q5 MCQ

Which one of the following geometric tolerances is always specified with respect to a datum?

(A) Flatness

(B) Straightness

(C) Circularity (roundness)

(D) Position

Q6 MCQ

In computer-aided process planning (CAPP), the variant (retrieval) approach

(A) creates every process plan from scratch using decision logic and a manufacturing database

(B) retrieves the standard plan of the part family (found by group-technology coding) and edits it for the new part

(C) needs no part classification or coding

(D) is used only for assembly planning

Q7 NAT

A fair die is thrown repeatedly until a six appears. The probability that more than four throws are needed (round off to three decimal places) is _____

Q8 NAT

The lengths of rods are normally distributed with mean 50 mm and standard deviation 4 mm. Given $\Phi(1) = 0.8413$ and $\Phi(0.5) = 0.6915$, the probability that a rod lies between 48 mm and 54 mm (round off to three decimal places) is _____

Q9 MCQ

If the correlation coefficient between x and y is zero, the two regression lines (y on x and x on y)

(A) coincide

(B) are parallel

(C) are perpendicular to each other, parallel to the coordinate axes

(D) make an angle of 45° with each other

Q10 NAT

The annual demand of an item is 7200 units, the ordering cost is ₹250 per order and the holding cost is ₹10 per unit per year. Under the EOQ policy, the number of orders placed per year is _____

Q11 MCQ

Increasing the number of periods n in a simple moving-average forecast

(A) makes the forecast respond faster to a change in demand

(B) makes the forecast smoother but slower to respond to a real change in demand

(C) gives more weight to the latest demand

(D) has no effect on the lag of the forecast

Q12 NAT

Five jobs with processing times of 5, 3, 8, 2 and 7 hours wait for a single machine and are sequenced by the SPT rule. The average number of jobs in the system over the makespan (round off to two decimal places) is _____

Q13 MCQ

In an activity-on-arrow project network, a dummy activity is used to

(A) show a correct precedence relationship or to keep two activities between the same pair of events distinct; it takes zero time and no resources

(B) represent an activity that is not on the critical path

(C) represent waiting time that consumes resources

(D) reduce the project duration by crashing

Q14 NAT

In an M/M/1 queue, the arrival rate is 4 per hour and the service rate is 5 per hour. The average number of customers waiting in the queue (excluding the one being served) is _____

Q15 MCQ

In the optimal simplex table of a maximisation LP, the shadow price (dual value) of a resource constraint means

(A) the cost of buying one unit of the resource in the market

(B) the number of units of the resource left unused

(C) the reduced cost of a non-basic variable

(D) the increase in the optimal objective value per unit increase in the availability of that resource, within the range of feasibility

Answer Key

Q	Type	Answer
Q1	NAT	7.13 (accept 7.08 to 7.18)
Q2	MCQ	(C) Selective laser sintering (SLS)
Q3	MCQ	(A) requires a backlash eliminator in the table feed screw because the cutter tends to pull the work into itself
Q4	NAT	1432.4 (accept 1425 to 1440)
Q5	MCQ	(D) Position
Q6	MCQ	(B) retrieves the standard plan of the part family (found by group-technology coding) and edits it for the new part
Q7	NAT	0.482 (accept 0.478 to 0.486)
Q8	NAT	0.533 (accept 0.529 to 0.537)
Q9	MCQ	(C) are perpendicular to each other, parallel to the coordinate axes
Q10	NAT	12
Q11	MCQ	(B) makes the forecast smoother but slower to respond to a real change in demand
Q12	NAT	2.36 (accept 2.34 to 2.38)
Q13	MCQ	(A) show a correct precedence relationship or to keep two activities between the same pair of events distinct; it takes zero time and no resources
Q14	NAT	3.2
Q15	MCQ	(D) the increase in the optimal objective value per unit increase in the availability of that resource, within the range of feasibility

Detailed Solutions

Q1 **Answer: 7.13 (accept 7.08 to 7.18)**

Concept: The compound rest is set to the half taper angle α, where $\tan\alpha = (D - d)/(2L)$.

$\tan\alpha = (60 - 50)/(2 \times 40) = 10/80 = 0.125$

$\alpha = \tan^{-1}(0.125) = \mathbf{7.13°}$

Exam tip: The compound-rest method suits short, steep tapers (hand feed); long, gentle tapers are turned by tailstock set-over or a taper-turning attachment.

Q2 **Answer: (C) Selective laser sintering (SLS)**

Concept: AM processes are grouped by the starting material: liquid (SLA: UV laser cures a photopolymer), solid filament/sheet (FDM extrudes, LOM cuts and bonds sheets) and powder (SLS, SLM, binder jetting).

A laser sintering a powder bed is **SLS (C)**; the unsintered powder supports the part, so no support structures are needed.

Exam tip: SLM/DMLS fully melts metal powder; SLS normally sinters polymers such as nylon.

Q3 **Answer: (A) requires a backlash eliminator in the table feed screw because the cutter tends to pull the work into itself**

Concept: In down milling the cutter rotates in the direction of the feed; each tooth enters at maximum chip thickness and exits at zero.

The cutting force pushes the work down onto the table and also in the feed direction, so any backlash in the lead screw lets the table jerk forward: a backlash eliminator is needed: **(A)**

Up milling starts at zero thickness (rubbing, poorer finish) and tends to lift the work, but it is used for hard-skinned castings.

Exam tip: Down milling gives better finish and longer tool life on rigid CNC machines with ball screws.

Q4 **Answer: 1432.4 (accept 1425 to 1440)**

Concept: With constant surface speed (G96), the control changes the spindle speed as the diameter changes: $N = 1000V/(\pi D)$.

$N = 1000 \times 180/(\pi \times 40) = 180\,000/125.66$

$N =$ **1432.4 rpm**

Exam tip: In facing towards the centre N rises sharply, so G50 (or G92) S... is used to clamp the maximum spindle speed; G97 cancels CSS.

Q5 **Answer: (D) Position**

Concept: Form tolerances (flatness, straightness, circularity, cylindricity) control a feature by itself and never use a datum.

Orientation (parallelism, perpendicularity, angularity), location (position, concentricity, symmetry) and run-out tolerances relate a feature to a datum: **(D)**

Exam tip: Profile of a line/surface may be used with or without a datum.

Q6 **Answer: (B) retrieves the standard plan of the part family (found by group-technology coding) and edits it for the new part**

Concept: CAPP has two approaches: variant and generative.

Variant: code the new part (GT), find its family, retrieve the family's standard plan and modify it: **(B)**

Generative: synthesises a new plan from part geometry, decision rules and process knowledge, with no stored family plan.

Exam tip: Variant systems are easier to build; generative systems are more consistent and need no existing plans.

Q7 **Answer: 0.482 (accept 0.478 to 0.486)**

Concept: More than k throws are needed exactly when the first k throws all fail: $P(N > k) = (1 - p)^k$.

$P(N > 4) = (5/6)^4 = 625/1296$

$=$ **0.482**

Exam tip: The expected number of throws is $1/p = 6$ (geometric distribution, which is memoryless).

Q8 **Answer: 0.533 (accept 0.529 to 0.537)**

Concept: Standardise: $z = (x - \mu)/\sigma$, and use $\Phi(-z) = 1 - \Phi(z)$.

$z_1 = (48 - 50)/4 = -0.5$; $z_2 = (54 - 50)/4 = 1$

$P = \Phi(1) - \Phi(-0.5) = 0.8413 - (1 - 0.6915) = \mathbf{0.533}$

Exam tip: Sketch the bell curve and shade the area; it avoids sign errors with negative z.

Q9 **Answer: (C) are perpendicular to each other, parallel to the coordinate axes**

Concept: The regression lines are $y - \bar{y} = r(\sigma_y/\sigma_x)(x - \bar{x})$ and $x - \bar{x} = r(\sigma_x/\sigma_y)(y - \bar{y})$.

With $r = 0$ they become $y = \bar{y}$ (horizontal) and $x = \bar{x}$ (vertical): perpendicular: **(C)**

With $r = \pm 1$ they coincide.

Exam tip: Both regression lines always pass through $(\bar{x}, \bar{y})$.

Q10 **Answer: 12**

Concept: $EOQ = \sqrt{(2DC_o/C_h)}$; number of orders = D/EOQ.

$EOQ = \sqrt{(2 \times 7200 \times 250/10)} = \sqrt{360\,000} = 600$ units

Orders per year = 7200/600 = **12**

Exam tip: Directly, $N^* = \sqrt{(DC_h/(2C_o))} = \sqrt{(7200 \times 10/500)} = 12$.

Q11 **Answer: (B) makes the forecast smoother but slower to respond to a real change in demand**

Concept: A simple moving average gives equal weight 1/n to each of the last n periods.

Larger n averages out random noise (smoother) but the forecast lags a trend or step change by about $(n - 1)/2$ periods: **(B)**

Exam tip: A moving average of n periods behaves roughly like exponential smoothing with $\alpha = 2/(n + 1)$.

Q12 **Answer: 2.36 (accept 2.34 to 2.38)**

Concept: Average number of jobs in the system = (sum of completion times)/(makespan), since each job stays in the system until it completes.

SPT order: 2, 3, 5, 7, 8 → completion times 2, 5, 10, 17, 25 (sum = 59)

Average number in system = 59/25 = **2.36**

Exam tip: Because the makespan is fixed, SPT, which minimises the sum of completion times, also minimises the average work-in-process.

Q13 **Answer: (A) show a correct precedence relationship or to keep two activities between the same pair of events distinct; it takes zero time and no resources**

Concept: In AOA networks each activity is an arrow between two events, and no two activities may share both end events.

A dummy (dashed arrow, zero duration, no resources) keeps parallel activities uniquely numbered and models logic that cannot be drawn otherwise: **(A)**

Exam tip: Activity-on-node networks need no dummies; that is one reason they are preferred in software.

Q14 **Answer: 3.2**

Concept: $L_q = \lambda^2/[\mu(\mu - \lambda)] = \rho^2/(1 - \rho)$.

$\rho = 4/5 = 0.8$

$L_q = 0.64/0.2 =$ **3.2** customers ($L = \rho/(1 - \rho) = 4$)

Exam tip: $W_q = L_q/\lambda = 0.8$ h by Little's law.

Q15 **Answer: (D) the increase in the optimal objective value per unit increase in the availability of that resource, within the range of feasibility**

Concept: The optimal dual variable $y_i = \partial Z^*/\partial b_i$; it is read from the net-evaluation row under the slack of constraint i.

It is the marginal worth of one more unit of the resource, valid while the current basis stays feasible: **(D)**

A resource with unused slack (non-binding constraint) has a shadow price of zero (complementary slackness).

Exam tip: Buy extra resource only if its market price is below its shadow price.

What next?

Day 66 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

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Next, Day 67/90: Engineering Mathematics, Maths Revision. Comment **GATE90** on our reel to get all links. Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.